Secondary math course practice

My role Supervising editor and SME review

I reviewed the math and teaching in a secondary teacher-preparation course, including practice on modeling real constraints with inequalities.

ORIGINAL DOCUMENT EXCERPT

Guided Practice

Multiple Select

A park is planning a tree-planting event and has two types of trees to plant: Oak trees (T) and Pine trees (P). The total number of trees planted cannot exceed 400. Additionally, each Oak tree requires 5 gallons of water per week, and each Pine tree requires 3 gallons of water per week. The park has a weekly water supply of 1,500 gallons for the trees.

Which of the following inequalities accurately represent the constraints in the tree-planting scenario? Select all that apply.

Correct answer: $T + P \leq 400$

Rationale: This inequality correctly represents the limit on the total number of trees that can be planted.

Correct answer: $5T + 3P \leq 1,500$

Rationale: This inequality correctly represents the limit on the amount of water available for the trees.

Distractor: $T + P \geq 400$

Rationale: This inequality incorrectly suggests that at least 400 trees must be planted, which is not stated in the problem.

Distractor: $5T + 3P \geq 1,500$

Rationale: This inequality incorrectly suggests that at least 1,500 gallons of water must be used, which is not stated in the problem.